

Asheesh Patel

Generative AI Engineer

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SUMMARY

GenAI Engineer with roughly 10 years of total professional experience, including 2+ years focused directly on AI/ML and Generative AI engineering. Experience spans backtesting and automating trading strategies through statistical and technical analysis, and developing/deploying GenAI (RAG) applications on AWS with LangChain, LangGraph, vector databases, and multi-agent orchestration. Comfortable designing layered retrieval architectures — e.g., moving from direct/structured lookup to constrained retrieval to full LLM-based generation — as demonstrated in a production Text-to-SQL system. Deploys models using a full MLOps stack (AWS, Git, Docker, DVC, MLflow) and builds guardrails for safe LLM outputs with Guardrails.ai. Currently expanding into graph-based retrieval methods (GraphRAG, Neo4j) to extend RAG systems beyond vector-only retrieval.

SKILLS

Programming: Python, SQL, NumPy, Pandas, Streamlit, Docker, PyTorch

Generative AI & LLM: LangChain, LangGraph, RAG, Retrieval-Augmented Generation, Prompt Engineering, Agentic AI, Multi-Agent Systems, OpenAI API, Hugging Face, Embeddings, Vector Databases, ChromaDB, Fine-tuning, PEFT, Function Calling, Tool Use, Langfuse, LangSmith, Guardrails

ML & MLOps: Machine Learning, NLP, Feature Engineering, MLflow, DVC, AWS (EC2, S3, Lambda), Git, CI/CD

Databases: PostgreSQL, Text-to-SQL, Schema-aware SQL Generation

Quant Finance: Value at Risk (VaR), Expected Shortfall, Monte Carlo Simulation, Backtesting, Algorithmic Trading

Currently Developing: Graph Databases (Neo4j), GraphRAG, Graph Algorithms for retrieval — self-directed study, applying graph-based retrieval concepts to existing RAG projects

PROFESSIONAL EXPERIENCE

Risk Analyst, *Entirety Markets, Noida, India* Jan 2024 – Feb 2026

- Designed and maintained Historical Simulation, Variance-Covariance, and Monte Carlo VaR models to quantify and monitor portfolio risk exposure across trading strategies.
- Built Python-based backtesting pipelines and portfolio performance monitoring tools; validated models using Kupiec's Proportion of Failures Test and Basel Traffic Light methodology.
- Developed a Generative AI system for financial news summarisation and market sentiment extraction using NLP and LLM models.
- Deployed analytics workflows on AWS EC2; used AWS S3 for scalable storage of historical datasets and model outputs.

Algorithmic Trading Strategy Developer, *Freelance* May 2022 – Dec 2023

- Developed and backtested algorithmic trading strategies using Python, Pandas, NumPy, Backtrader, and Zipline, focusing on signal quality and risk-adjusted returns.
- Designed statistical models and technical indicators to improve trade signals.

Mathematics Teacher, *Vikas Vidya Mandir, Kanpur* Aug 2016 – Apr 2022

- Delivered mathematics and analytical problem-solving training for competitive examinations.
- Mentored students in quantitative reasoning and logical problem solving.

PROJECTS

Text2SQL — Natural Language to SQL Query Generator

Python, LangChain, OpenAI, PostgreSQL, Langfuse, Docker | github.com/ASHEESHPEL1996/Text2SQL

- Architected a production-grade Text-to-SQL system with a tiered resolution flow — schema-aware direct lookups first, falling back to constrained, validated SQL generation via LangChain/OpenAI only when needed — enabling safe, read-only natural-language querying of a PostgreSQL database.

- Reduced LLM inference latency by approximately 90%, significantly lowering costs and boosting system efficiency.
- Built a Langfuse observability pipeline for real-time tracking of token usage, latency, errors, and cost; containerized with Docker and deployed on Streamlit Cloud with query history and CSV export.

AI Code Documentation Generator

Python, OpenAI API, AST, Streamlit | github.com/ASHEESH Patel1996/code-generation

- Built a GenAI application that automatically generates structured docstrings and project overviews for entire codebases — reducing documentation time from hours to seconds.
- Implemented an AST-based chunking algorithm (2,000-line segments, 50-line overlap); reduced API costs by 90% via caching and token-efficient prompt design.
- Designed an AST-based safe insertion mechanism to validate semantic equivalence, preventing unintended logic changes across file, ZIP, and GitHub repository inputs.

AI Investment Research

Python, LangChain, LangGraph, LLAMA | github.com/ASHEESH Patel1996/ai-finance-analyst

- Developed a production-ready, multi-agent investment research platform using LangGraph and Groq (Llama 3.1), with autonomous, role-specialized agents — Stock Analyst, Portfolio Risk, and Sentiment Tracker — coordinated through a defined agent-orchestration pattern to deliver institutional-quality research.
- Developed and integrated financial data pipelines from multiple sources (yFinance, Alpha Vantage, Yahoo Finance RSS, Google News RSS) for real-time market intelligence, including sentiment classification and deduplication for data quality.
- Designed and implemented a full-stack AI application with a FastAPI backend and Streamlit frontend, using a modular prompt-engineering approach across all agent workflows.

EDUCATION

Bachelor of Science — Mathematics | Chhatrapati Shahu Ji Maharaj University, Kanpur | 2013 – 2016

Bachelor of Education | Prof. Rajendra Singh (Rajju Bhaiya) University, Prayagraj | 2022 – 2024

CERTIFICATIONS

- Python Data Fundamentals — DataCamp
- The Complete SQL Bootcamp — Udemy
- Safe and Reliable AI via Guardrails — DeepLearning.ai